Multiscale hybrid modeling framework for intercrops : the best of crop and plant models

PhD in computer science, 2nd year

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Intercropping

= agroecological pratice where at least two crop species are grown together

(Vandermeer, 1989)

Potential benefits :

- more resilience to stresses
- reduction of use of fertilizers and pesticides
- increased yield

(Jensen et al., 2020, Stomph et al., 2020, Gaudio et al., 2021)

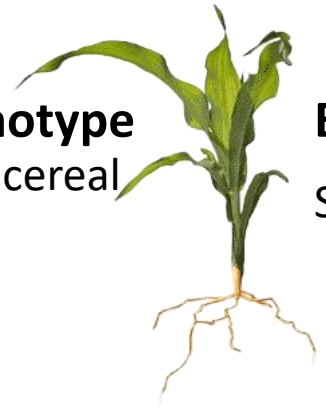
… only if intercropping systems are optimized

Intercropping systems

G x G x E x M

Genotype
of a legume

Genotype
of a cereal



Environnement

Soil & atmosphere

Management

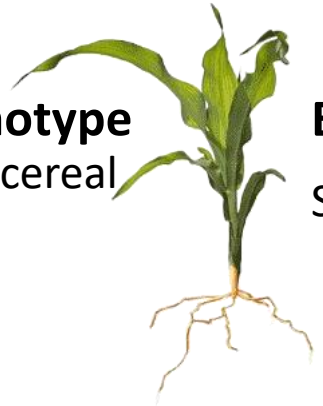
Sowing pattern

Inputs

Intercropping systems

G x G x E x M

Genotype
of a legume



Genotype
of a cereal

Environnement

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Inputs

Interspecific interactions

Heterogeneous spatial and temporal distribution of resources

Intercropping systems

G x G x E x M

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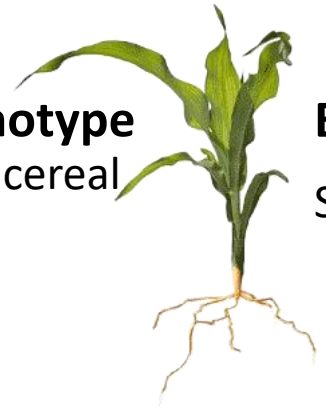

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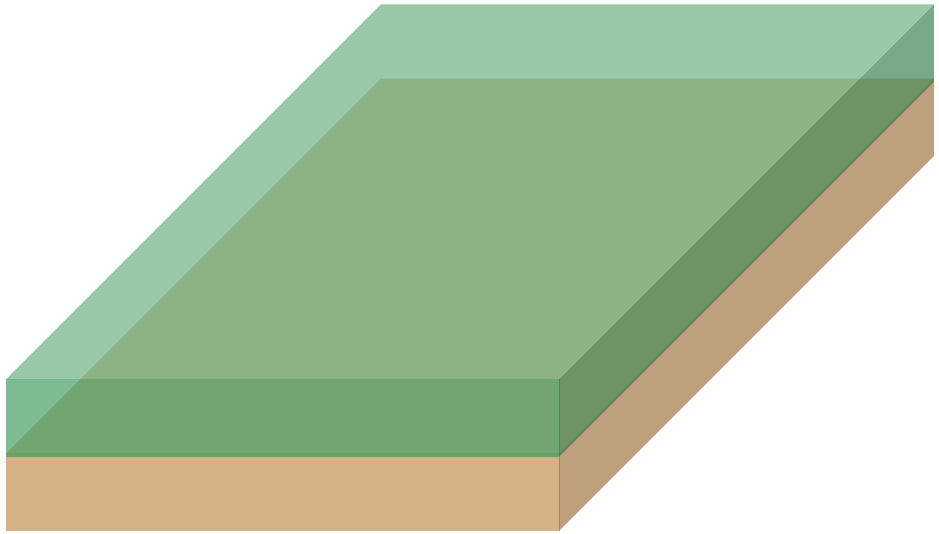
➔ **Crop and FSP models** are needed to **explore the complexity** of these systems **at different scales.**

Crop models

(Boote et al., 2013, Jones et al., 2017)
e.g.: STICS (Vezy et al., 2023),
APSIM (Githui et al., 2023)



<u>Objective:</u> prediction, decision-support tool

<u>Characteristics:</u>

- crop scale
- homogeneous hypotheses
- simulate multiple processes through the whole development cycle
- already calibrated for intercrops

<u>Limits:</u>

- no heterogeneity
- no explicit plant-plant interactions

FSPM (Functional-Structural Plant Models)

(Evers et al., 2019 , Louarn and Song, 2020)
e.g.: Virtual GrassLand (Louarn et al., 2020),
SIMBAL (Perthame et al., 2023)

<u>Objective:</u> integration of knowledge at a organ scale, plasticity studies

<u>Characteristics:</u>
- organ scale
- 3D plant structure influence and emerges from plant processes
- enable to represent heterogeneous crops

<u>Limits:</u>
- complexity
- cannot always simulate whole development cycle
- need lots of 3D measures

How to benefit from the best of both approaches?

➔ Possible way: by **adapting crop models** for considering more **architectural and spatial heterogeneity** of intercrops (Gaudio et al., 2019; Laurans et al., 2024)

1. Evaluate performance of existing crop model formalisms for heterogeneous canopies

Turbid medium approach for simulating light interception
(Barillot et al., 2011; Feng et al., 2014; Ponce de Leon and Bailey, 2019; Li et al., 2024)

2. Identify architectural traits of interest for a given process

Impact of plant architectural variations on light interception
(Barillot et al., 2014; Truong et al, 2015; Barillot et al., 2019)

3. Find new formalisms for crop models

Hybrid 2.5D formalism in STICS
(Vezy et al., 2023)

Incorporate effect of clumping
(Bailey et al., 2020)

My PhD : Multiscale hybrid modelling for intercrops

For given processes, what in plant architecture should we consider in intercrop models?

Chapter I

ArchiCrop,
a 3D dynamic parametric model

*How to conceive a **3D plant model** with **few architectural and developmental parameters** and with a canopy-scale **constraint on growth** ?*

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Uncertainty analysis of intercrop models considering architectural parameters

*Which **plant architectural traits** and **sowing pattern** bring most of the **uncertainty** and should be considered explicitly in crop models?*

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Chapter III

Proposition of multi-scale modelling approaches

*How to **consider explicitly** identified parameters in **crop models** in order to improve their accuracy?*

SPPM: Structural Parametric Plant Model

- Plant **3D architecture** and **development** are defined by **parameters**.

≠ FSPM: 3D plant architecture emerges from plant processes

- Few parameters → known **allometric laws** for potential plant

- Explore morphological space

SPPM: Structural Parametric Plant Model

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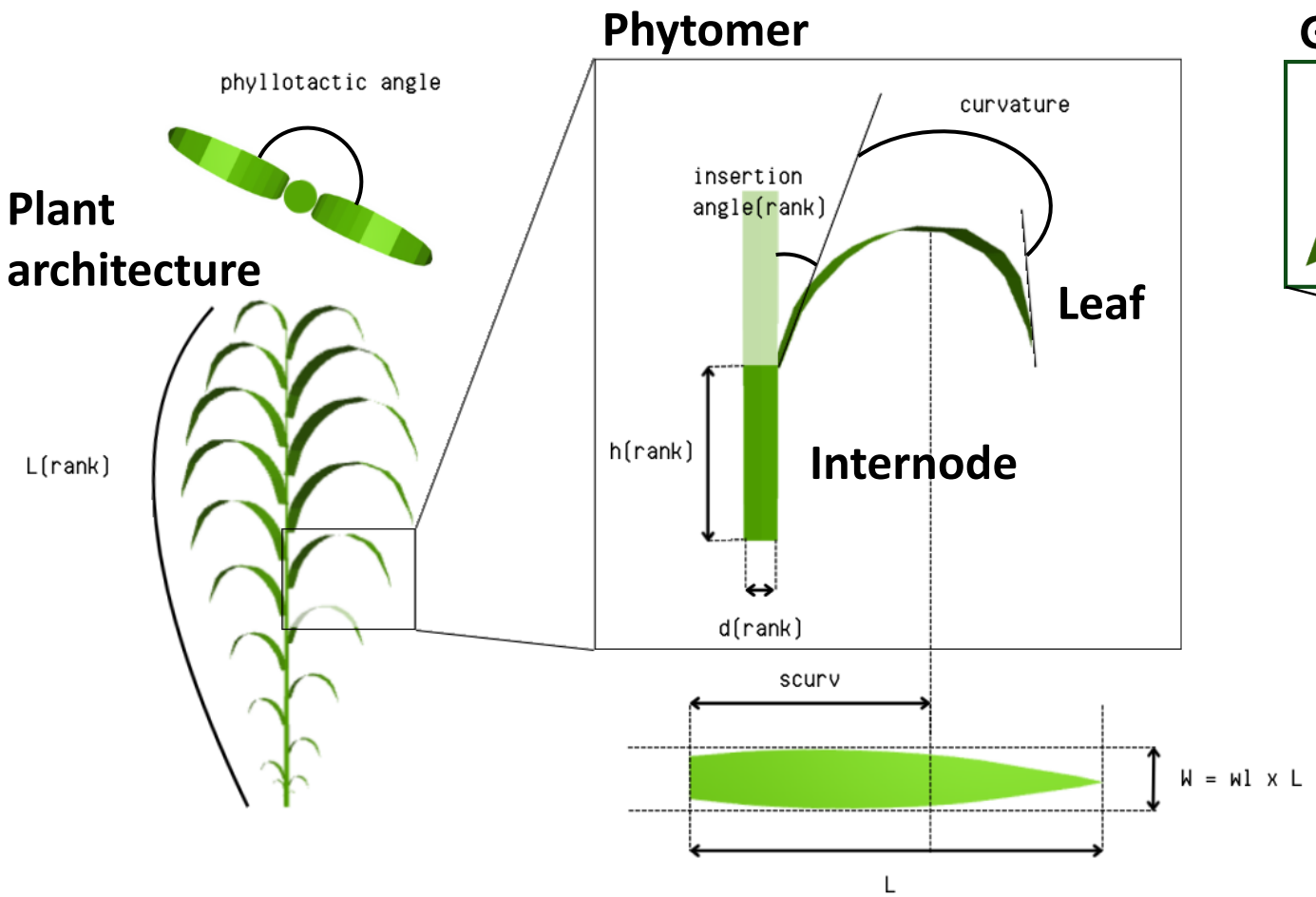
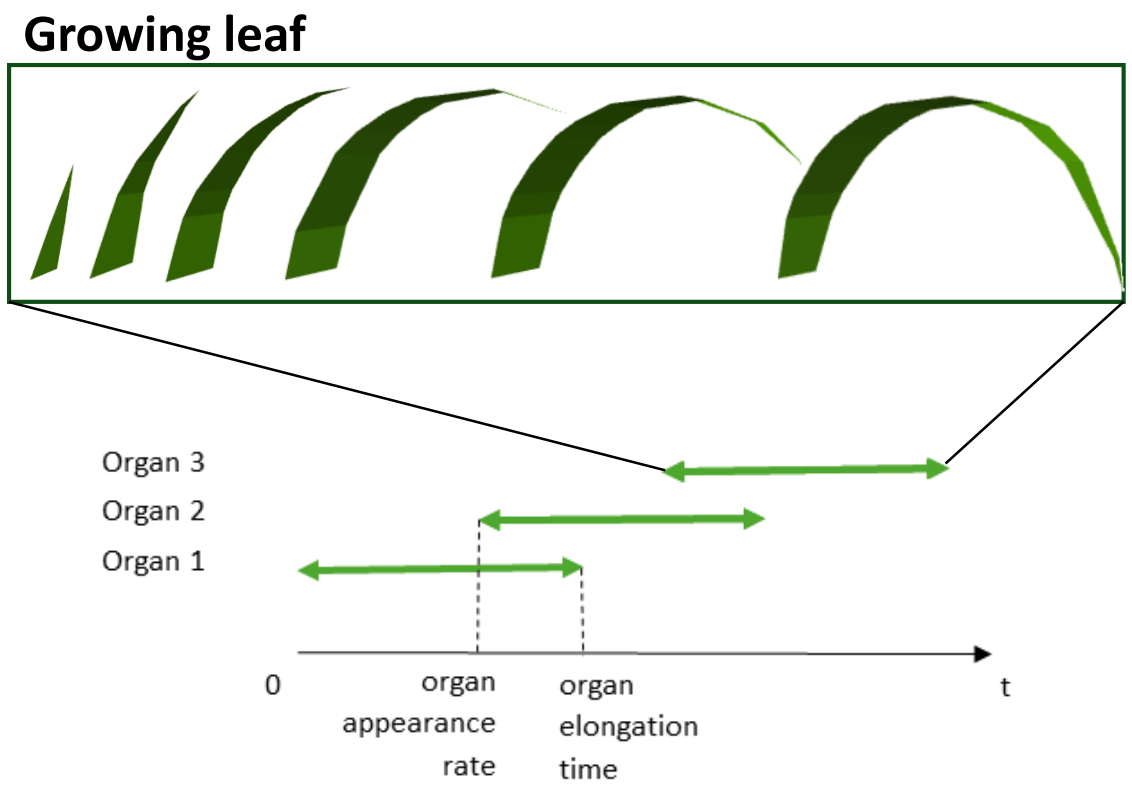
ArchiCrop: a SPPM for cereals

→ where **growth** is constrained from **crop model dynamics**

Potential plant architecture and development are defined by parameters.

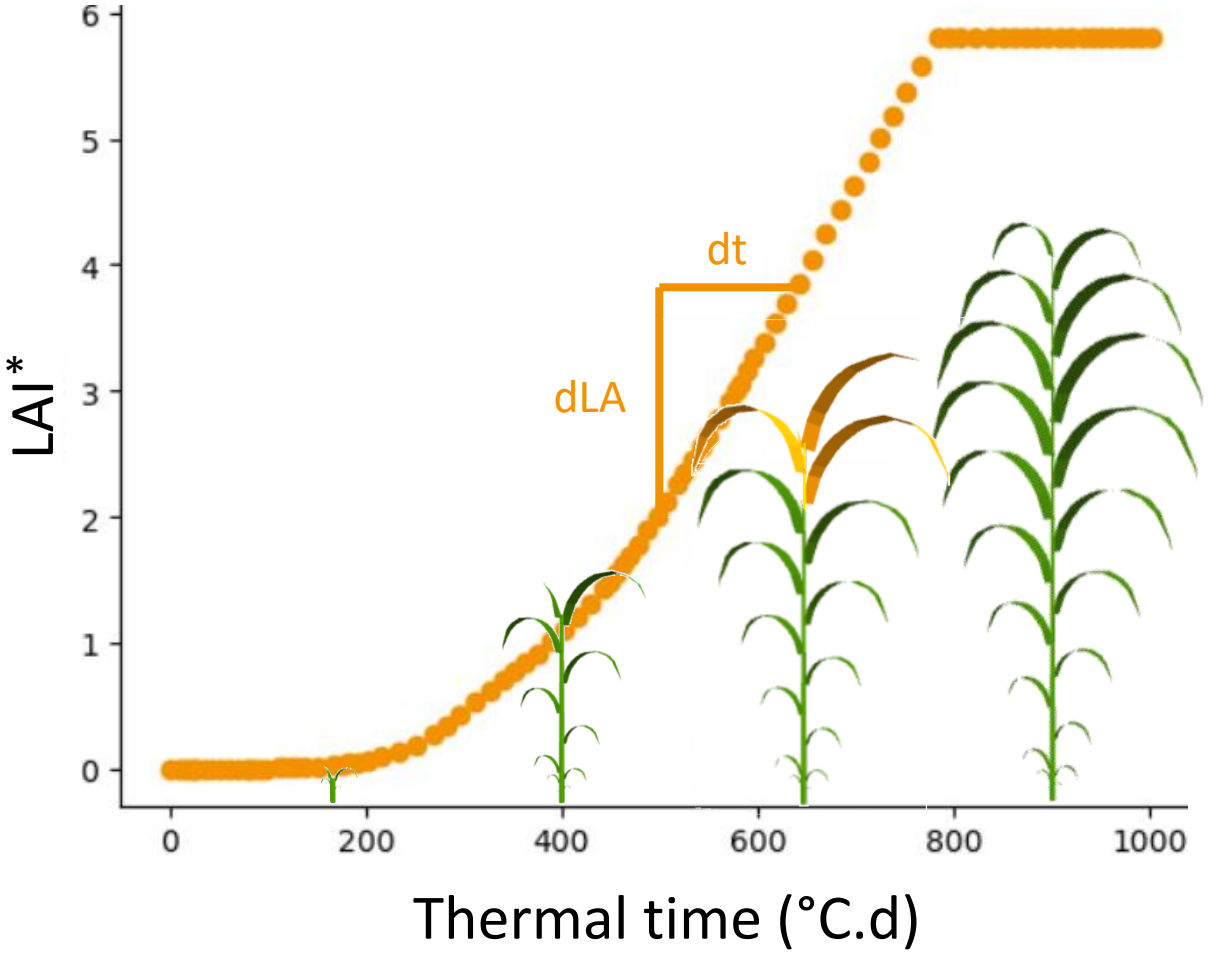


Geometrical parameters



Developmental parameters

Organ growth is constrained by canopy-scale dynamics generated by crop model.



*LAI = Leaf Area Index

How to **distribute a constraint *c*** (LA and height) among *n* growing elements ?

→ **among plants** in a canopy and **among organs** within a plant

$$\frac{\partial c}{\partial t} = \sum_{i \in G} \sum_{j \in P} \frac{\partial c_{i,j}(t)}{\partial t}$$

$\frac{\partial c}{\partial t}$: growth increment at crop scale

$\frac{\partial c_{i,j}(t)}{\partial t}$: growth increment for an organ

P: set of plants of the same species in a crop

G: set of growing organs in a plant

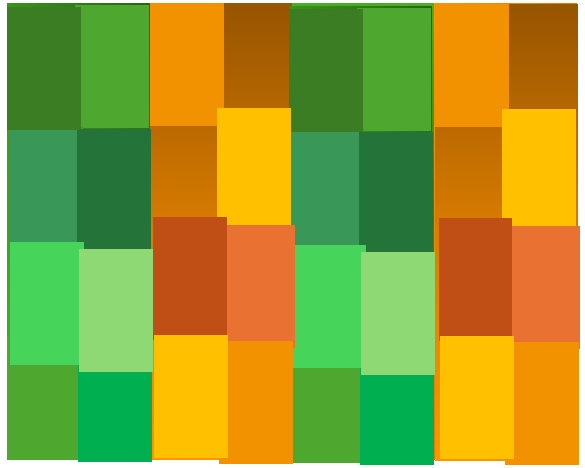
Different scales of heterogeneity for an intercrop

Interspecific heterogeneity

→ Intercropping sowing pattern

Intraspecific heterogeneity

→ Varietal variability



Inter-organs heterogeneity

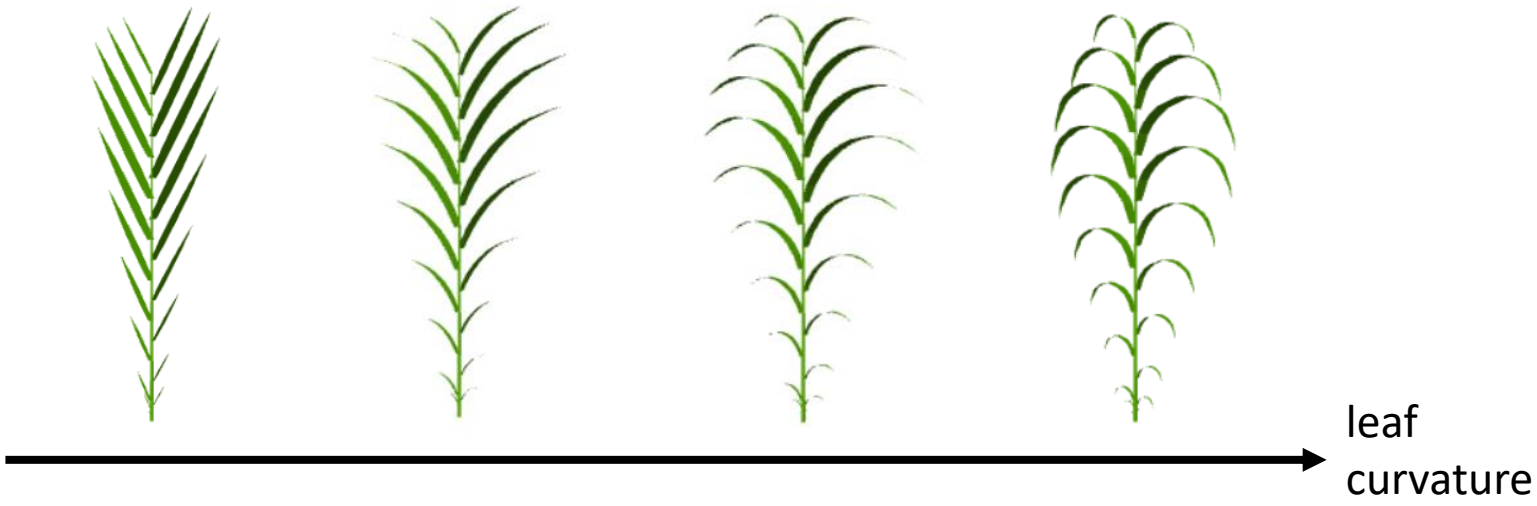
→ Plant architecture

→ **degrees of freedom** compared to crop model

Create morphospace of potential plants

Architectural degrees of freedom:

- number of phytomers
- longest leaf length
- leaf width-to-length ratio
- internode lengths
- insertion angle
- leaf curvature



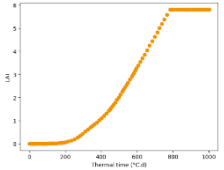
Workflow of simulations of intercrops through time

<u>ExM experimental design</u>

x

1 simulation with crop model

Growth dynamics LAI(t), height(t)



ArchiCrop

<u>GxG experimental design :</u>
Morphospace species 1
x
Morphospace species 2
x
Spatial configurations

Generation of 3D scenes of intercrops through time

OpenAlea

Workflow of simulations of intercrops through time

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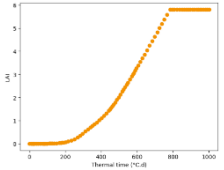

ArchiCrop

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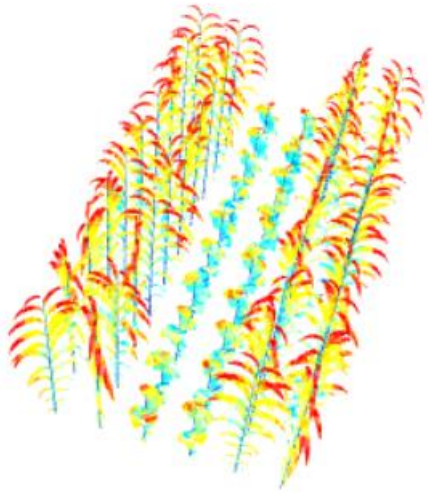
Computation of light interception at organ scale



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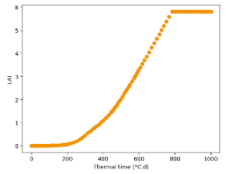

ArchiCrop

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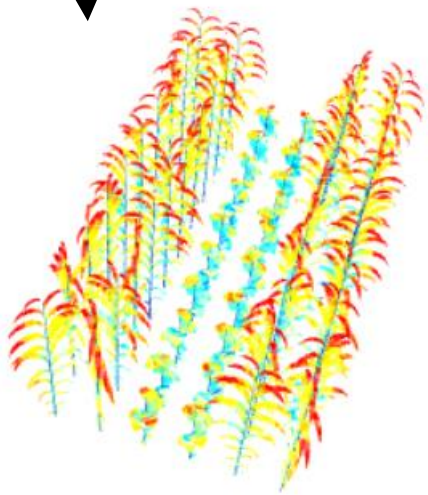
Analysis of the uncertainty of the crop model associated to architecture parameters and spatial configuration



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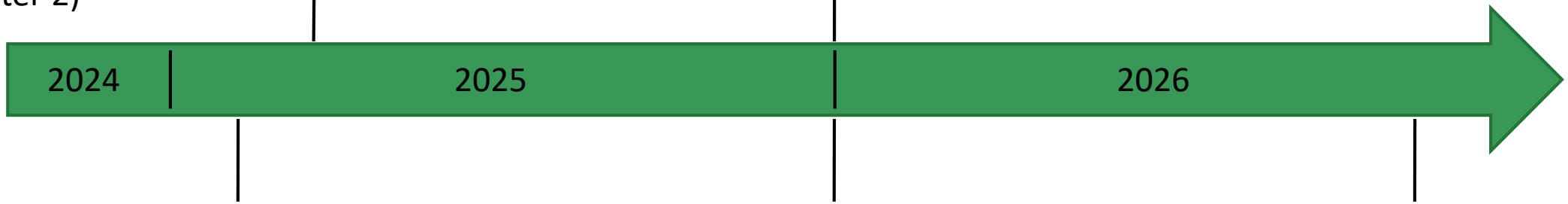
Perspectives

- Extension of the cereal model to legumes (chapter 1)
- Simulations and uncertainty analyses for light interception (chapter 2)

Similar approach for roots, looking at processes like water and nitrogen uptake (chapter 1&2)

Propositions of new formalisms for STICS crop model (chapter 3)



First article on chapter 1 submitted

Second article on chapter 2 submitted

Thesis manuscript ready